

6. This question is about the uses of gases present in air.

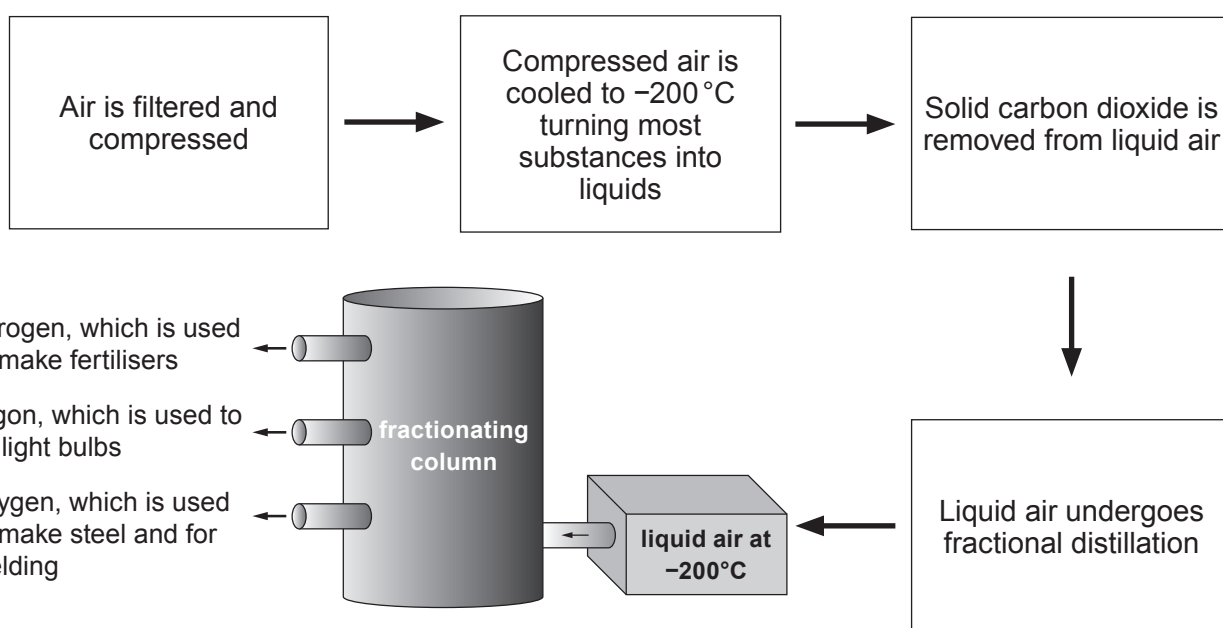
Air is a mixture of different gases. Each gas has different uses. The gases are present in different amounts as shown below.

| Gas              | Amount present in air | Boiling point (°C) |
|------------------|-----------------------|--------------------|
| N <sub>2</sub>   | 78.04 %               | -195.8             |
| O <sub>2</sub>   | 20.95 %               | -183.0             |
| CO <sub>2</sub>  | 0.03 %                | -78.5              |
| Ar               | 0.93 %                | -189.2             |
| Ne               | 18 ppm                | -246.0             |
| He               | 5 ppm                 | -268.9             |
| Kr               | 1 ppm                 | -152.3             |
| Xe               | 0.08 ppm              | -107.1             |
| H <sub>2</sub>   | 0.5 ppm               | -257.9             |
| CH <sub>4</sub>  | 2 ppm                 | -164.0             |
| N <sub>2</sub> O | 0.5 ppm               | -88.5              |

ppm = parts per million

Separating a complex mixture!

Air is firstly compressed and cooled which turns it into a liquid. Carbon dioxide freezes and is removed as solid dry ice. The rest of the mixture goes on to a fractionating column where it is slowly warmed allowing most of the substances to be separated.



- (a) Tick (✓) the **main** reason why hydrogen is not separated during the fractional distillation of liquid air. [1]

hydrogen is a highly reactive gas

☐

only 0.5 ppm of hydrogen is present

☐

hydrogen does not become liquid on cooling to  $-200^{\circ}\text{C}$

☐

hydrogen has a higher boiling point than helium

☐

- (b) Tick (✓) the reason why carbon dioxide becomes a solid during the first part of the process. [1]

carbon dioxide has a boiling point above  $-200^{\circ}\text{C}$

☐

carbon dioxide has a melting point above  $-200^{\circ}\text{C}$

☐

carbon dioxide has a melting point below  $-200^{\circ}\text{C}$

☐

carbon dioxide has a boiling point below  $-200^{\circ}\text{C}$

☐

- (c) Describe, in terms of boiling points, how nitrogen, argon and oxygen are separated during fractional distillation. [3]

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Examiner  
only

(d) The noble gases are present in very small quantities in the air. The amount of each gas produced per year is shown in the table.

| Noble gas | World production (tonnes per year) |
|-----------|------------------------------------|
| helium    | 28 000                             |
| neon      | 1 000                              |
| argon     | 700 000                            |
| krypton   | 8                                  |
| xenon     | 0.6                                |

Use the data to calculate the number of tonnes of air needed to produce 700 000 tonnes of argon. Give your answer in **standard form**. [2]

Mass of air needed = ..... tonnes

|   |
|---|
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